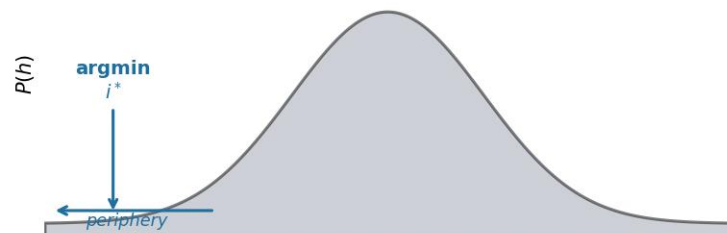


Two BIB lineages: argmin replacement vs argmax modification

Argmin lineage [Gunji 2018, 2021; Ibuka & Sasai 2024; this work]
— inverse step acts on the **lowest**-posterior hypothesis

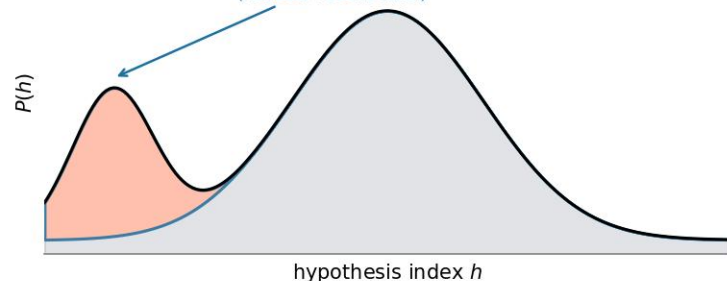
before

$$L_{i^*} \leftarrow \text{REPLACE with empirical } p^t$$



after

new hypothesis
(reflects recent data)

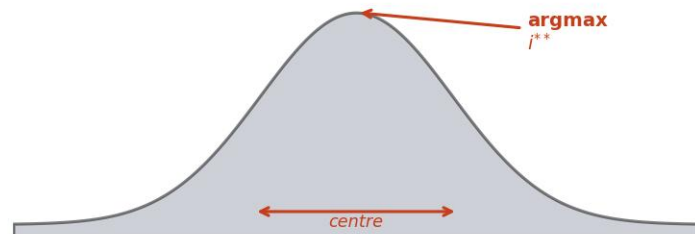


change at the **periphery**: new hypothesis injected at low- P region

Argmax lineage [Shinohara 2020, 2021, 2022]
— inverse step acts on the **highest**-posterior hypothesis

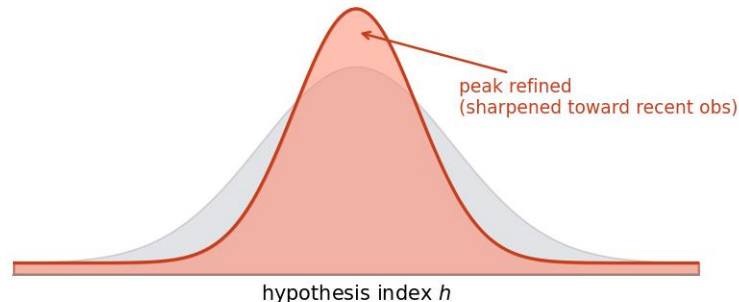
before

$$L_{i^{**}} \leftarrow \text{REFINE by mixing in latest observation}$$



after

peak refined
(sharpened toward recent obs)



change at the **centre**: dominant hypothesis refined in place

Both lineages produce heavy-tailed persistence statistics, but the relaxation channel acts at different locations in the hypothesis space — periphery (argmin) vs centre (argmax). Whether the two share a single universality class is an open empirical question.